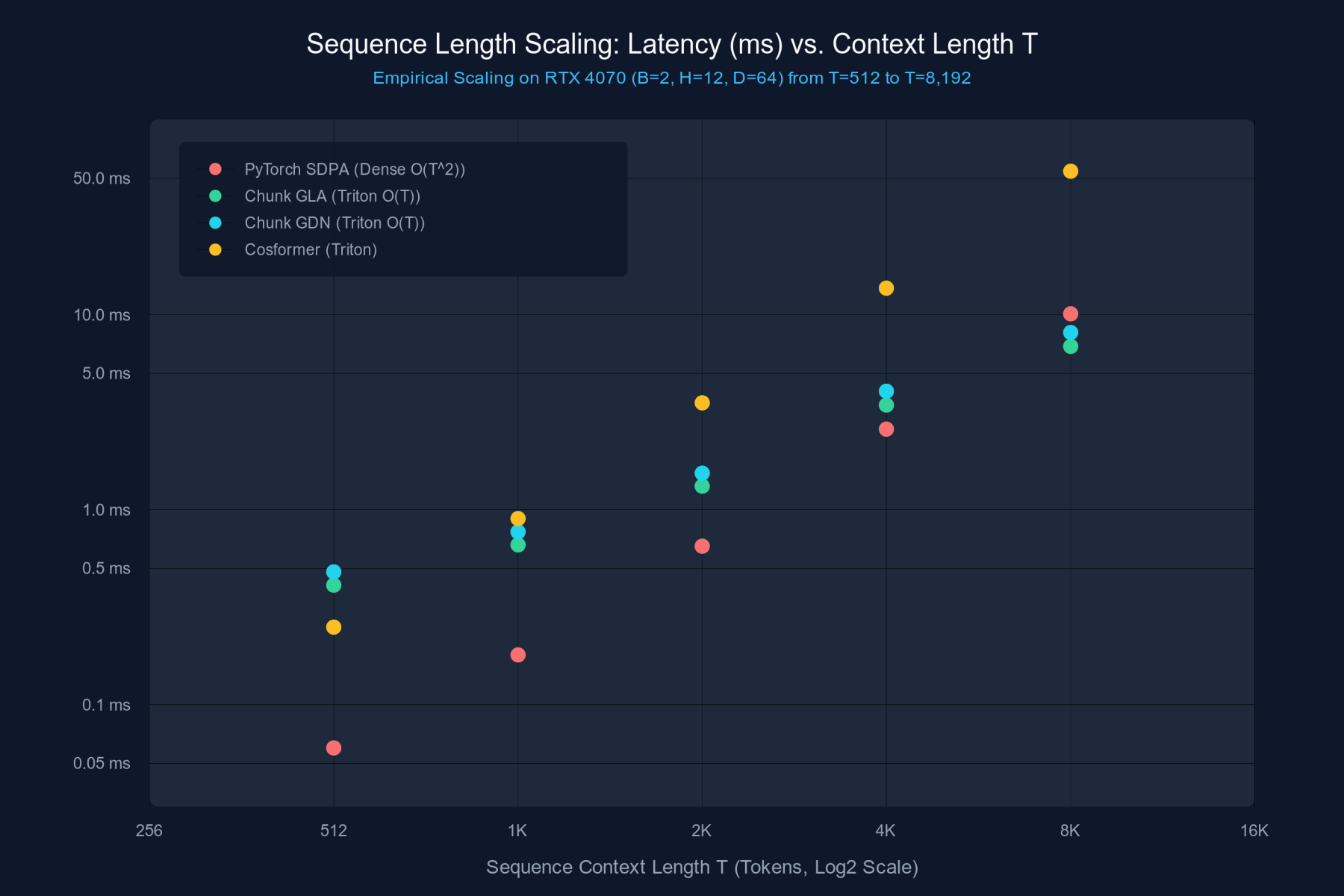

Sequence Length Scaling: Latency (ms) vs. Context Length T
Empirical Scaling on RTX 4070 (B=2, H=12, D=64) from T=512 to T=8,192
PyTorch SDPA (Dense O(T^2))
Chunk GLA (Triton O(T))
Chunk GDN (Triton O(T))
Cosformer (Triton)
50.0 ms
10.0 ms
5.0 ms
1.0 ms
0.5 ms
0.1 ms
0.05 ms
256
512
1K
2K
4K
8K
16K
Sequence Context Length T (Tokens, Log2 Scale)